KRUSHI RAKSHAK

AI-Enabled Field Protector

1. Problem Statement

Farmers cultivating agricultural land near forests and wildlife habitats frequently face crop damage caused by wild animals such as wild boars, monkeys, and elephants. These animals can enter farms unexpectedly, damage crops, and cause financial losses. Farmers often depend on manual monitoring, especially at night, which is difficult to sustain continuously. Krushi Rakshak addresses the need for early detection and timely warning of animal intrusion.

2. Need for the Innovation

There is a need for an affordable, sustainable, and practical method to monitor agricultural fields and identify wildlife intrusion at an early stage. The solution should reduce dependence on continuous manual monitoring and be suitable for locations where reliable electricity may be limited.

3. Objectives

- Detect animal intrusion in agricultural fields using a camera.
- Use AI/ML-based image recognition to identify targeted animals.
- Provide an early warning when an identified animal is detected.
- Use solar power to support operation in remote agricultural areas.
- Develop a practical solution that can be adapted to different farms and wildlife-related challenges.

4. Solution / Innovation Description

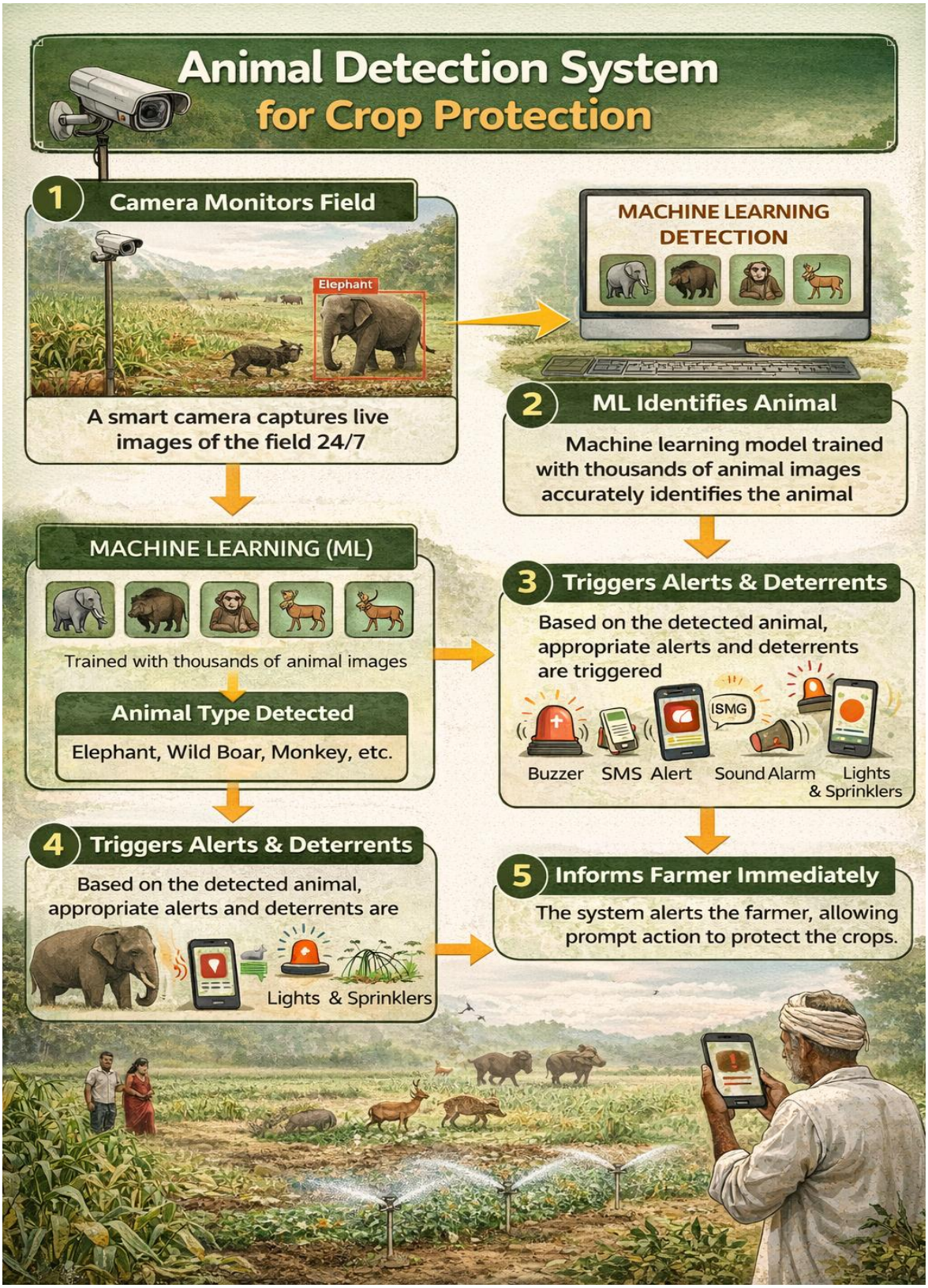
Krushi Rakshak (Field Protector) is an AI-enabled, solar-powered animal intrusion detection and alert system designed to help farmers protect their crops from wildlife.

The system uses a camera to monitor agricultural fields and capture images when animals enter the monitored area. An AI/ML-based animal detection model analyzes the captured images and identifies animals such as wild boars, monkeys, and elephants.

Once an animal is detected, the system provides an early warning alert, enabling farmers to take timely action to protect their crops. The system is powered by a solar panel, making it suitable for agricultural areas where access to reliable electricity may be limited.

5. How the System Works

The working of the system is shown in the image below:



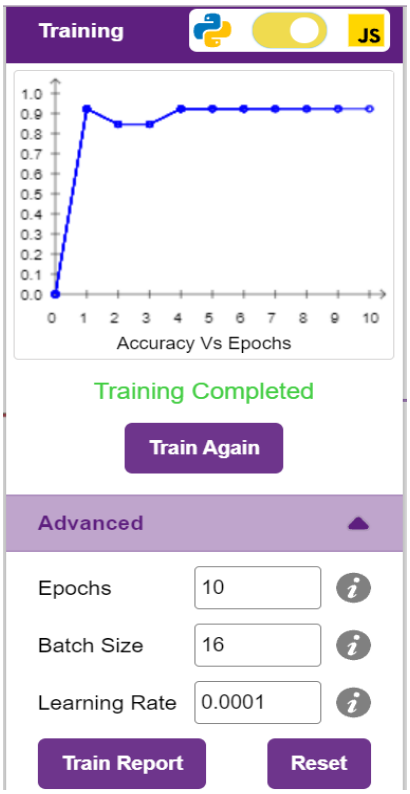
Step	Stage	Description
1	Field Monitoring	A camera monitors the agricultural field and captures relevant images.
2	Image Analysis	The captured image is processed using an AI/ML-based animal detection model.
3	Animal Identification	The model identifies targeted animals such as wild boar, monkey, or elephant.
4	Early Warning	When an animal is detected, the system provides an alert to support timely action.
5	Solar Power	A solar panel supports operation in locations with limited access to electricity.

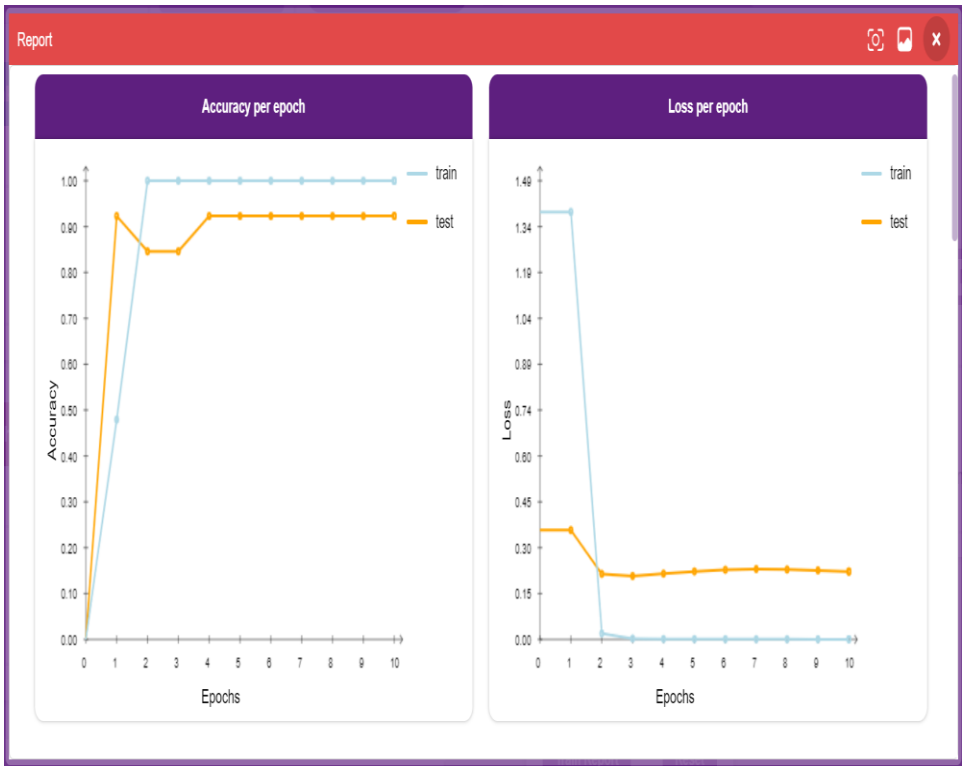
6. Technology and Components

- Camera – captures images for animal detection.
- AI/ML model – identifies targeted animals from captured images.
- Solar panel – provides renewable power for field deployment.
- Alert mechanism – provides an early warning when animal intrusion is detected.
- AI/ML tools and model development – Pictoblox, Keras-based approach, with model training supported by image datasets.

7. AI/ML Model

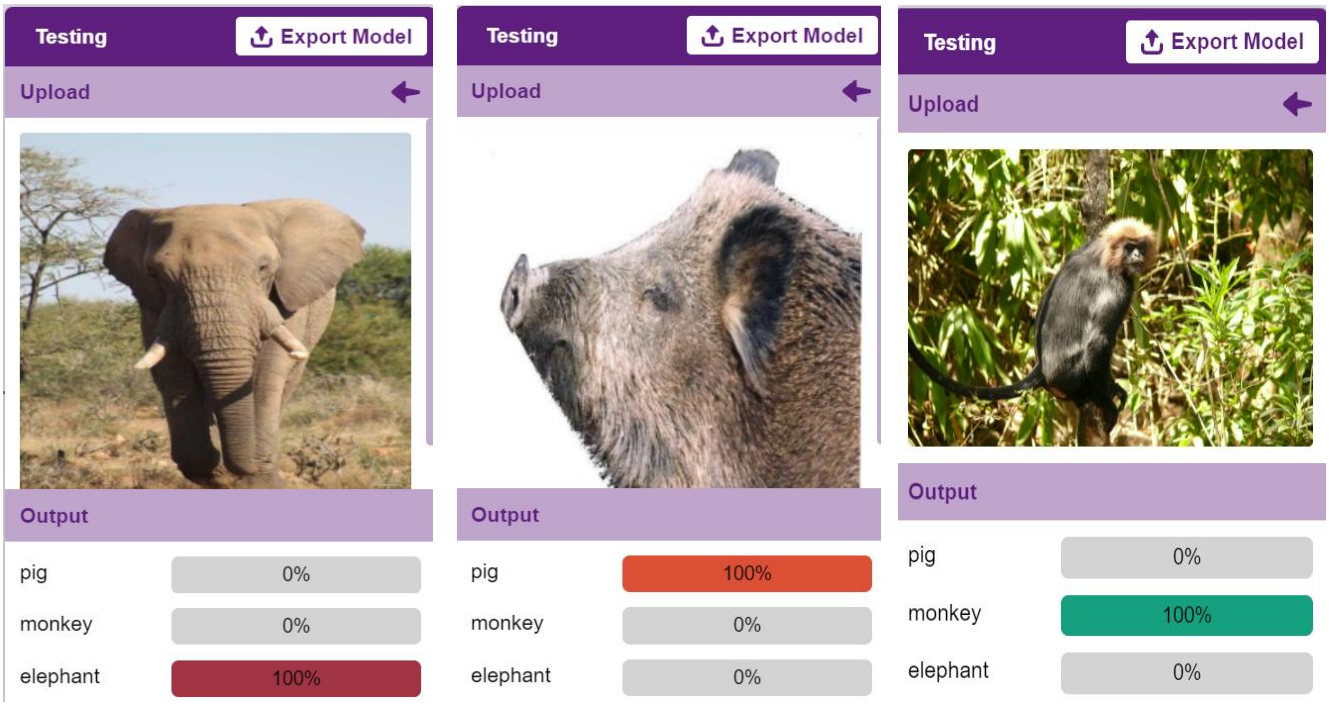
The project uses an AI/ML-based image classification/detection approach trained to recognize target animal categories. The intended categories include wild boar, monkey, and elephant. The model is designed to support early identification of animal intrusion rather than relying only on generic motion detection.

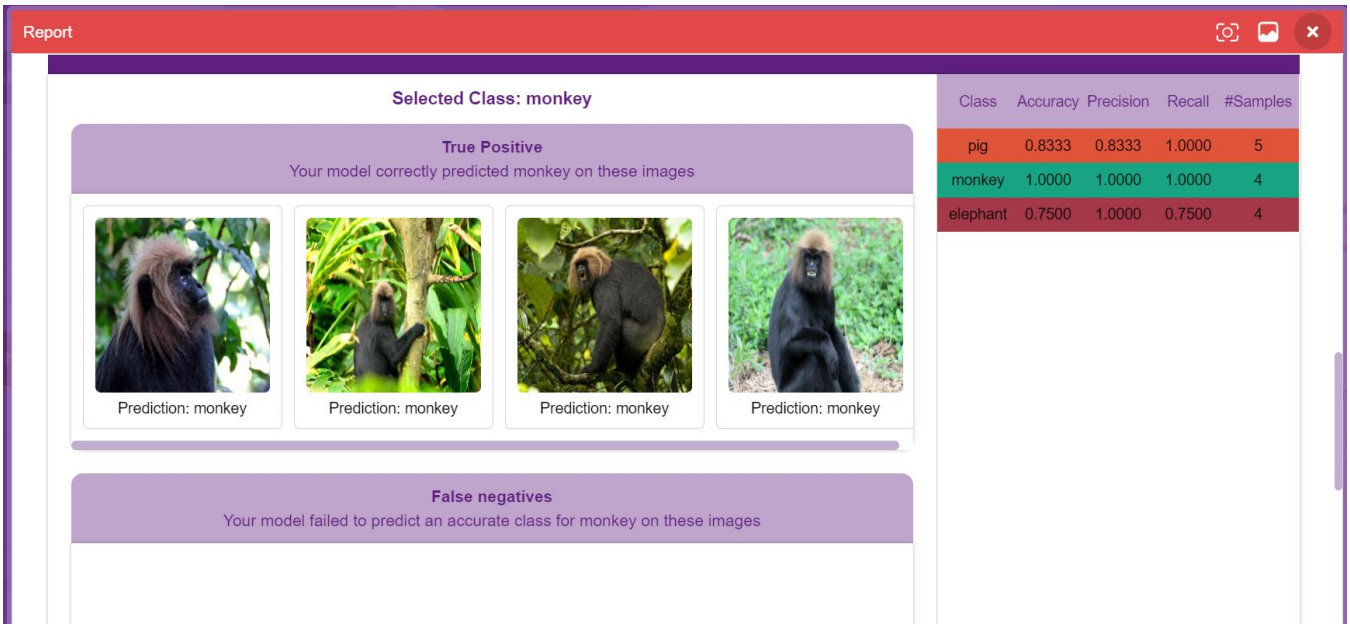




8. Testing Methodology

The prototype should be evaluated using representative images or field conditions for each target animal. Testing should record whether the animal was correctly identified, whether an alert was generated, and the response time. Tests should also consider different lighting and field conditions.





9. Testing Results

The testing-results table is intentionally left for actual prototype observations. No accuracy or performance figures are claimed here without recorded test evidence. The final submission can be updated with the number of trials, correct detections, missed detections, false detections, alert response time, and solar-operation observations.

10. Expected / Observed Benefits

- Reduces the need for continuous manual field monitoring.
- Supports early identification of animal intrusion.
- Helps farmers respond in a timely manner.
- Uses solar power, supporting deployment in remote areas.
- Can potentially be adapted to different crops, locations, and animal categories.

11. What Makes the Idea New or Different?

Krushi Rakshak combines AI-based animal identification, camera monitoring, automated alerts, and solar power in one practical solution. Instead of only detecting movement, the approach aims to identify the type of animal entering the field, helping farmers make a more informed and timely response.

12. Beneficiaries

The primary beneficiaries are farmers, especially small and marginal farmers whose agricultural fields are located near forest boundaries, wildlife habitats, or animal movement zones. Farming communities and rural communities affected by human-wildlife interactions may also benefit.

13. Where and How It Can Be Used

Krushi Rakshak can be deployed in agricultural fields near forests, wildlife corridors, forest boundaries, and areas frequently affected by animal intrusion. It can be installed at field boundaries or vulnerable points for day and night monitoring. Its solar-powered design makes it suitable for remote locations with limited access to reliable electricity.

14. Impact

- Potential reduction in crop losses caused by unexpected wildlife intrusion.
- Improved ability of farmers to receive early warnings.
- Reduced dependence on continuous manual night-time monitoring.
- Promotion of affordable and sustainable technology for rural agricultural applications.
- Potential contribution to safer coexistence between farming communities and wildlife through early detection.

15. Limitations

- Detection performance depends on the quality and diversity of the training dataset.
- Lighting, weather, distance, occlusion, and camera placement may affect detection.
- The system provides an alert and does not by itself guarantee prevention of animal entry or crop damage.

- Further field testing is required to establish performance across different farms and conditions.

16. Future Scope

- Expand the AI model to recognize additional wildlife species.
- Improve performance under low-light, rain, and other challenging field conditions.
- Add multiple cameras for wider field coverage.
- Develop improved alert and communication options for farmers.
- Conduct longer-duration field trials to measure accuracy, reliability, and crop-loss reduction.
- Explore integration with a farmer-friendly mobile or web interface.

17. SDG Alignment

The project can contribute to Sustainable Development Goals related to Zero Hunger (SDG 2), Industry, Innovation and Infrastructure (SDG 9), Responsible Consumption and Production (SDG 12), Climate Action (SDG 13), and Life on Land (SDG 15), particularly through technology-enabled crop protection, renewable energy use, innovation, and human-wildlife coexistence.

18. Conclusion

Krushi Rakshak is a practical student-led innovation that applies AI/ML, camera-based monitoring, automated alerting, and solar power to a real agricultural problem. By enabling early detection of wildlife intrusion, the project aims to help farmers protect crops, reduce dependence on continuous manual monitoring, and support more sustainable agricultural practices.